

SparkValidate

Risk-based validation and alarm rationalization for Ignition and PLC systems, delivered as an Ignition gateway module plus a standalone CLI (Validate Desk).

Version 0.1.0 Ignition 8.1.19+ Runs on: any gateway, or CLI-only offline

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1. Overview

SparkValidate is a deterministic validation scaffold for Ignition and PLC systems. It normalizes a system model, applies a risk-policy YAML, and generates a Validation Master Plan (VMP) as Markdown and DOCX. The gateway module adds live gateway extraction and protocol execution against a running system; the CLI (Validate Desk) runs standalone for offline or CI use and does not require a gateway.

SparkValidate is not a general document-management system and does not replace a quality system. It generates VMP documents, IQ evidence, and a requirements traceability matrix (RTM) from a system model and risk policy you author; it does not author your URS or your risk policy for you.

Extension seams (`io.sparkvalidate.ingest.Ingestor`, `io.sparkvalidate.risk.RiskEngine`, `io.sparkvalidate.docs.DocGenerator`) are stable interfaces for ingest, risk-scoring, and document generation. The Phase 0 VMP generator is the first concrete `DocGenerator`.

2. What You Need

- An Ignition gateway for the gateway module (live extraction, protocol execution, gateway REST routes).
- Java 17 to build or run the CLI jar; the gateway module targets Java 11 bytecode. The CLI ships as a self-contained jar (`cli-0.1.0-all.jar`) so an end user only needs a Java 17 runtime, not the Gradle toolchain.
- No other SparkBridge module is required. The gateway module and Validate Desk CLI are licensed and sold separately.

3. Install

Gateway module

1. Download the signed SparkValidate .modl from the release page.
2. Gateway web UI > Config > Modules > Install or Upgrade a Module, upload the file, accept the EULA (see `license.html` at the repository root for full terms).

3. If the module is self-signed, accept the untrusted-certificate warning to complete install.
4. No build step is required for the release bundle.

Validate Desk CLI

1. Download the CLI release (self-contained jar) or build it: `./gradlew clean build`, producing `cli/build/libs/cli-0.1.0-all.jar`.
2. Start a validation repository: `java -jar cli-0.1.0-all.jar init validation/`. This creates the plan layout: `model/system.json`, `policy/risk.yaml`, `requirements/urs.yaml`, `rationalization/mad.yaml`, plus `docs/`, `protocols/`, `evidence/.gitattributes`, and `rtm/rtm.csv`.

4. License

Gateway module: licensed per gateway, perpetual. Drop `sparkbridge-license.key` in the gateway's data directory; the module re-reads it within 30 seconds. See `license.html` at the repository root for the full EULA.

Validate Desk (CLI) carries its own separate license, sold per user, standalone from the gateway module.

A free tier provides the alarm KPI report and rationalization gap list only, with no document generation or protocol execution.

5. Configure

Gateway config tab

Setting	Meaning
enabled	Turns the module's live extraction and protocol execution on or off.
snapshotdir	Directory where config snapshots (with hashes) are written for IQ evidence.
journalprofile	Journal/history profile used when a protocol needs to reference historical data.
alarmprovider	Alarm provider read for alarm KPI reporting and rationalization.
simprovider	Simulation provider used when protocol execution needs to write simulated values.
apitokenadminrole	Gateway role required to issue REST API bearer tokens.
allowsimwrites	Permits protocol execution to write to the simulation provider.
allowinsecureloopback	Allows unauthenticated loopback access to gateway REST routes; off by default.

CLI: generating a VMP

```
java -jar cli/build/libs/cli-0.1.0-all.jar generate \  
  --model model.json \  
  --policy policy/risk.yaml \  
  --out out/
```

Writes `out/docs/VMP.md` and `out/docs/VMP.docx`. The Markdown renderer is canonical and deterministic for a given model, policy, and injected clock; the DOCX renderer uses Apache POI.

Gateway REST authentication

Routes under `/data/io.sparkvalidate` require gateway authentication: either a bearer token (issued via a session-cookie-authenticated tokens issue call) or an authenticated session cookie.

6. Operate

Day to day, SparkValidate is driven from the CLI (generating or regenerating the VMP as the model or policy changes) and from the gateway module when protocols run against a live system for IQ evidence. Watch the gateway REST status route for module health and licensing state. Config snapshots taken for IQ evidence are written to snapshotdir; keep that directory backed up alongside the rest of the validation repository.

7. Verify

- CLI: sparkvalidate gateway status reads gateway or module status with a bearer token.
- Gateway: GET /data/io.sparkvalidate/status reports module and gateway status, including enabled and allowInsecureLoopback.
- Confirm a VMP generation run produces both out/docs/VMP.md and out/docs/VMP.docx with no errors.

8. Troubleshoot

Symptom	Cause	Fix
Install warns about an untrusted certificate	Module is self-signed	Accept or trust the certificate to complete install
/data/io.sparkvalidate/status returns unauthorized	No bearer token or session cookie supplied	Issue a token via a session-cookie-authenticated tokens issue call, or authenticate the session first
Protocol execution cannot write simulated values	allowsimwrites is off	Enable allowsimwrites in the gateway config tab
DOCX generation fails while Markdown succeeds	Apache POI rendering issue with the current model or policy	Check the CLI output for the POI error; regenerate after fixing the offending model or policy field
Only KPI report and gap list are available, no document generation	Running unlicensed (free tier)	License the gateway module or Validate Desk CLI

9. Reference

Item	Value
Gateway REST base	/data/io.sparkvalidate
License key path	<data>/sparkbridge-license.key, re-read within 30 s
CLI jar	cli/build/libs/cli-0.1.0-all.jar
Init layout	model/system.json, policy/risk.yaml, requirements/urs.yaml, rationalization/mad.yaml, docs/, protocols/, evidence/.gitattributes, rtm/rtm.csv
Extension interfaces	io.sparkvalidate.ingest.Ingestor, io.sparkvalidate.risk.RiskEngine, io.sparkvalidate.docs.DocGenerator
Toolchain	Gradle 8.10.2, Java 17 to run Gradle, Java 11 bytecode for core/gateway, Java 17 for the CLI

10. Support

[SparkValidate install and config reference](#) · [Contact](#) · [Downloads and release notes](#)